PYCO: A Physical Coherence Token Emerging from the Lindblad Protocol

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Abstract

PYCO is the native token of the Lindblad Protocol. It was not designed as a financial instrument — it emerges as a necessary consequence of a network that measures and rewards physical coherence. This paper describes the mechanism by which PYCO is generated, distributed, and consumed within the Spectral Ledger, and establishes the economic properties that result from anchoring token issuance in physical hardware validation rather than computational work or economic stake. Every PYCO in existence was produced by a physical node running the Lindblad Cryptography Protocol (LCP) stack on real hardware. As of June 2026, over 1,512,000 PYCO have been mined across 35,842+ epochs by physical hardware nodes deployed on mainnet.

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1 Origin — Why a Token Emerges

The Lindblad Protocol operates a distributed network of physical hardware nodes. Each node continuously performs four-layer physical verification (LCP L1–L4): silicon identity via SRAM PUF, cryptographic signing via P-256 ECDSA, temporal entropy via the Hybrid Stochastic Chua circuit (HSC), and irreversible consensus via the Lindblad master equation over a LoRa mesh network [4].

This verification work is real, measurable, and continuous. The network requires a mechanism to:

1. **Quantify** the physical coherence contributed by each node
2. **Reward** nodes proportionally to their verified contribution
3. **Enable** value transfer out of the Spectral Ledger into public blockchains

PYCO fulfills all three functions. It is the unit of account for physical coherence accumulated in the Spectral Ledger.

PYCO is issued exclusively through physical mining by hardware nodes running the LCP stack. The token supply is a direct record of physical work performed on the network since genesis.

2 Generation — Physical Coherence Verification (PCV-4)

2.1 The Mining Mechanism

PYCO is generated through Physical Coherence Verification (PCV-4). Each epoch (≈60 seconds), active nodes submit a block containing:

- A hardware-signed attestation (L1 + L2)
- A temporal proof from the Chua HSC (L3)
- A consensus digest from the LoRa mesh (L4)

The Spectral Ledger fullnode validates each attestation and credits the node with a base reward:

$$r_{\text{base}} = 4.275 \text{ PYCO per epoch per active node} \tag{1}$$

This reward accumulates in the node's balance within the Spectral Ledger and becomes transferable upon claim to the node's LD-address.

2.2 The Physical Coherence Verification Score (PCV-4)

The PCV-4 score is a composite measure of node quality across four dimensions:

Table 1: PCV-4 Score Dimensions

Dimension	Source	Measures
U — Uptime	Epoch participation ratio	Fraction of epochs with valid block submission
S — Spectral consistency	PUF signature stability	Stability of silicon fingerprint over time
L — Latency	Block submission speed	Time from epoch start to block arrival
E — Entropy quality	Chua HSC output	Statistical quality of physical entropy source

The PCV-4 score governs each node's share of bridge fee rewards:

$$\text{reward}_i = F \times \frac{\text{PCV4}_i}{\sum_j \text{PCV4}_j} \tag{2}$$

where F is the total fee pool available for distribution in a given epoch.

2.3 Supply Cap

The maximum supply of PYCO is **100,000,000 PYCO**, fixed in the ERC-20 contract deployed on Arbitrum One:

`0x16a69CcdA3865a23537d46055dC6564A2813C36B`

No additional issuance is possible beyond this cap. Mining continues until the cap is reached, at which point nodes are compensated exclusively through bridge fee distribution.

3 The Spectral Ledger — Internal Economy

3.1 Architecture

The Spectral Ledger is the internal state layer of the Lindblad Protocol. It operates as a gasless, instant settlement layer for PYCO, USDT, USDC, and future tokens. All internal transfers between LD-addresses are free and finalized within the next epoch.

Each participant on the Spectral Ledger is identified by an LD-address derived from the node's SRAM PUF:

```
Full PUF:      484C59B737BB65B2       (16 hex chars, from SRAM PUF)
Short ID:      LD484C5                (LD + first 7 chars)
Full address: LD-484C59B737BB65B2   (used in API calls)
```

The LD-address is physically unique — derived from the SRAM PUF of the hardware at boot, stabilized by the BCH(255, 139, t=15) fuzzy extractor [4], and registered in the Spectral Ledger upon first successful block submission.

3.2 Internal Transfers

Transfers between LD-addresses on the Spectral Ledger are gasless, instant (finalized within the current epoch), and irreversible. The state transition for an internal transfer is:

$$\begin{aligned} \text{balance(sender)} &\leftarrow \text{balance(sender)} - \text{amount} \\ \text{balance(recipient)} &\leftarrow \text{balance(recipient)} + \text{amount} \\ \text{nonce(sender)} &\leftarrow \text{nonce(sender)} + 1 \end{aligned} \tag{3}$$

The updated state is included in the next epoch block and propagates across all nodes via the LoRa mesh.

4 Bridge Exit — The Consumption Mechanism

4.1 Hardware-Validated Withdrawal

Exiting the Spectral Ledger to a public blockchain requires a hardware-signed withdrawal request. The process:

1. User requests withdrawal via LindWallet
2. Fullnode generates a cryptographic challenge for the paired node
3. Node signs the challenge with its PUF-derived ECDSA key (L1 + L2)
4. Signature is verified on-chain via the bridge smart contract
5. Tokens are released on the destination chain

Every withdrawal is signed by physical silicon — the hardware that generated the LD-address must be present and operational to authorize the exit.

4.2 The Bridge Fee

Each withdrawal incurs a fee of **0.1%** of the withdrawn amount, denominated in PYCO:

$$f = w \times 0.001 \text{ (in PYCO)} \tag{4}$$

where w is the withdrawal amount. This fee is split:

Table 2: Bridge Fee Distribution

Destination	Amount	Effect
Burn (zero address)	50% of fee	Permanent supply reduction
Active node operators	50% of fee	Distributed by PCV-4 score

4.3 Supported Networks

Table 3: Bridge-Supported Networks

Network	Chain ID	Tokens
Arbitrum One	42161	USDT, USDC, PYCO
Polygon	137	USDT, USDC

PYCO exists on Arbitrum One only and functions as the native token of the entire network regardless of which chain a bridge deposit originates from.

5 Economic Properties

5.1 Supply Dynamics

The PYCO circulating supply follows:

$$S(t) = \min\left(\int_0^t m(\tau)\, d\tau,\ 10^8\right) - B(t) \tag{5}$$

where $m(t)$ is the instantaneous mining rate (PYCO per epoch per active node) and $B(t)$ is the cumulative burned supply. The cap of 10^8 PYCO is absolute. The circulating supply decreases monotonically once bridge volume generates sufficient burn.

5.2 Demand Structure

Demand for PYCO arises from two sources:

Functional demand: Every withdrawal from the Spectral Ledger requires PYCO to pay the bridge fee. Users without PYCO must acquire it before withdrawing — creating continuous, usage-driven demand proportional to network transaction volume.

Node operator supply: Operators running physical nodes accumulate PYCO through mining rewards. Operators may hold, sell, or use PYCO within the ecosystem, creating natural market depth.

5.3 The Physical Coherence Economy

The PCV-4 score creates an economic incentive structure tied directly to hardware performance. A node with higher uptime, more stable PUF signature, faster block submission, and higher entropy quality receives a proportionally larger share of the fee pool. This incentive structure aligns economic reward with physical network quality — operators are compensated for running reliable, high-quality hardware.

6 Token Distribution

PYCO has no pre-mine, no team allocation, no investor round, and no foundation reserve:

Table 4: PYCO Token Distribution

Source	Allocation	Mechanism
Physical mining	100% of supply	Earned by hardware nodes via PCV-4
Pre-mine	0%	None
Team	0%	None
Investors	0%	None
Foundation	0%	None

Every PYCO in existence was produced by a physical node running the LCP stack on real hardware. The distribution of PYCO across addresses is a direct record of physical work performed on the network since genesis.

7 Relationship to the Lindblad Master Equation

The Spectral Ledger state — including all PYCO balances — evolves as a density matrix ρ governed by the Lindblad master equation [1, 2]:

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right) \tag{6}$$

Each PYCO balance is the expectation value of an observable operator over the current density matrix:

$$\langle \mathrm{Balance}_{\mathrm{addr}} \rangle = \mathrm{Tr}(O_{\mathrm{addr}} \cdot \rho) \tag{7}$$

The dissipative term in Eq. (6) ensures that state transitions — including token transfers and burn events — are thermodynamically irreversible. A completed transfer cannot be reversed without violating the second law of thermodynamics. This gives PYCO a property not found in software-based token systems: transaction finality is a consequence of thermodynamic law.

8 Live Network Data (June 2026)

Table 5: Live Network Statistics — June 2026

Metric	Value
PYCO mined (total)	>1,512,000 PYCO
PYCO burned	0 PYCO
Active mining nodes	3 physical hardware nodes
Current epoch	35,842+
Base reward per epoch	4.275 PYCO per node
Max supply	100,000,000 PYCO
Contract (Arbitrum One)	`0x16a69CcdA3865a23537d46055dC6564A2813C36B`
Bridge networks	Arbitrum One + Polygon

References

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Lindblad Protocol — The hardware decides. The physics guarantees. The chain records.